

Atmospheror: Towards an Olfactory Interactive System for Enhancing Social Presence and Interaction in Synchronous Online Classes

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Figure 1: The Atmospheror system for olfactory interaction in a synchronous online classroom setting. a) The instructor receives "Thumbs up" feedback from students in the form of lemon scents. b) A student presses a button based on how she is feeling about the lecture. c) A student receives the scent of mint when the instructor presses the "Request attention" button.

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ABSTRACT

Synchronous online classes have become increasingly common in recent years. However, students and instructors often face challenges when giving and receiving reactions and feedback. Existing solutions include audio and visual cues, which often act as distractions. A potential alternative is olfactory cues. This paper proposes Atmospheror, an interactive system that supports interaction and feedback between students and instructors through an ambient olfactory display. We performed a pilot study with two instructors and 13 students to evaluate the effectiveness of Atmospheror and its design and implementation, applying both quantitative and qualitative methods. Preliminary results indicate that Atmospheror

can promote students' concentration and increase interactivity in online classes. User interviews provided suggestions for future enhancements of the selection and treatment of odor, hardware implementation, and interface system mechanisms.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices; Interactive systems and tools; User studies.**

KEYWORDS

olfactory interface, synchronous online class, scent atmosphere, social presence and interaction, remote communication

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1 INTRODUCTION

Synchronous learning refers to real-time interaction between geographically distanced instructors and learners using multimedia tools in a live virtual environment [5, 24]. The COVID-19 pandemic has led to a rapid increase in the usage of synchronous online education. However, it still faces many challenges in strengthening the connection between instructors and students [11, 12, 19, 45]. Compared to in-person classes, the lack of social presence and interaction during online classes remains one of the most significant problems [11, 19]. For in-person classes, personal response systems can enhance active social interaction [9, 10, 18, 36, 41, 42]. However, it is worth noting that all these efforts usually convey information through visual and audio cues. Hence, extra visual and audio cues may impose an additional cognitive load and distraction for both the presenter and the audience in online synchronous classes [36, 42].

To address this issue, introducing a new sensory channel into the interaction may be a viable solution. Since Kaye proposed the idea of symbolic aroma output [23], there have been many studies on using olfactory displays for message notifications [3, 16, 29]. Furthermore, ambient display techniques are often used to provide social presence and a sense of awareness in group learning environments [1, 4]. Studies have also shown that smell can be used to deliver messages during visual overload [22, 37]. Further, odor-cued memories are more detailed and emotional than verbal-cued memories [15], and since both emotion and memory have a significant impact on student and teacher status [7, 34, 39], this inspired researchers to use olfactory cues in interface design.

In this paper, we demonstrate a new possibility of using olfactory interfaces to assist synchronous online classrooms in order to enhance the sense of presence and interaction. We designed an olfactory interactive system "Atmospheror" that is used by all students and instructors in a virtual classroom. The system simultaneously releases odors depending on overall student status determined using anonymous, crowd feedback strategies. We conducted a pilot user study to explore perceptions of experiencing synchronous

online classes when assisted with Atmospheror, as well as preferences for information conveyance settings. Based on the results, the system is initially validated for its potential to increase classroom interaction, as well as its ability to positively affect the teaching of instructors and the learning moods of students. Section 5 contains a detailed description of statistical results, interview findings, and limitations. Section 6 draws conclusions and discusses the direction of our future work.

2 RELATED WORK

2.1 Challenges and Efforts for Synchronous Online Classes

Early research in this area focused on the design and development of synchronous online learning environments [6]. Recently, COVID-19 accelerated the popularity of synchronous online learning [12, 17, 20]. However, Yarmand et al. reminds us of the interaction gaps in synchronous online classrooms for teachers, students, and peers [11]. There have been several solutions that were developed recently to reduce the difficulty of monitoring audience reactions in online classrooms, such as dynamically spotlighting expressive audience members [33], and detecting learners' statuses through algorithms [27]. However, audience visibility, while allowing the instructor to be aware of student status through their expressions, also leads to distractions [30] as well as privacy issues on the student side [27]. Chen et al. analyzed the live-streaming-based remote education experience in Chinese colleges [12], and their distilled design implications including "diverse interaction formats" and "balance of anonymity and real-name system" inspired us to include related aspects into our design.

The quality of online education also depends on innovations in communication and interaction methods, and when social presence in online classrooms is increased, student motivation, engagement, and perceived learning are enhanced [32, 43]. One type of innovation is in personal response systems (self-reporting systems, clickers) [9, 35], which can promote classroom interaction and cognitive engagement [31, 44], and has proven effective in enhancing social presence, interaction, and engagement [36]. For example, Teevan et al. developed a smartphone-based application for audiences to give thumbs up/thumbs down in real-time [42]. Glassman et al. designed a prototype named Mudslide, in which students can give feedback to teachers [18]. In comparison, we combined self-report mechanisms with web-connected olfactory ambient displays [22] to encourage in-class interaction while maintaining minimal distractions and protecting privacy.

2.2 Ambient Displays and Olfactory Interfaces for Promoting Social Awareness and Interaction

There are various ambient display systems [43] that have been developed to increase user awareness of their environmental changes [4]. In terms of visual and auditory modalities, Brewer et al. designed an interface to increase workgroup awareness, which could capture and express the activities of co-workers [8]. An et al. presented

ClassBeacons, which uses lamps to visualize how a teacher distributes their time and attention in a classroom [1]. In comparison, scent is less disruptive than visual and auditory modalities when providing notifications [2, 3]. Additionally, olfaction can help users perceive information when they are experiencing visual overload [37]. Thus, using scent to convey information in online education can reduce the visual cognitive load on teachers and students.

Moreover, the olfactory sense is directly connected to emotional and memory processing areas in the brain [21], and as such olfactory ambient displays could facilitate social interaction by evoking emotions and memory [14, 26]. Obrist et al. suggested utilizing smell to trigger discussions and to provide pleasant reminders. Xiang et al. proposed a set of odor "emoticons" to represent and share moods [44]. Moreover, Choi et al. found that pleasant aromas could enhance student mood and learning [13]. All of these advantages can be used to enhance engagement and social connection in synchronous online classes.

3 ATMOSPHEROR

In order to explore whether olfactory interaction notifications increase student interactivity, engagement, and concentration in online classrooms, we developed an Atmospheror System (Fig. 1) based on an ambient olfactory display for the following reasons: 1) As a new interaction channel, olfaction can enhance the multimodal properties of the classroom without increasing audio or visual cognitive load. 2) From a physiological and psychological perspective, scents have the ability to influence and guide the statuses of students. 3) Olfaction is closely related to long-term memory, and hence we expect that olfactory interactions can deepen student and instructor impressions of lecture details.

3.1 Olfactory Ambient Display System

The Atmospheror system includes student devices, instructor devices, and client software. The system supports the remote release of different scents controlled by users. Through the release of scents, students can give feedback to the teacher on their learning status, and the teacher can also guide student statuses (Fig. 2). The devices have different scent channels, as well as different buttons that are used to send states to the client software (Fig. 3). The Atmospheror client software runs in the background and is responsible for communication with the hardware and the server, as well as displaying the thresholds settings for scent delivery (available only to instructors). A server receives data from the distributed clients and is responsible for tracking the number of students with each status, and then triggering relevant responses in client devices to output a unified odor diffusion signal (see Fig. 3).

In terms of odor source selection, we referred to previous psychological studies regarding the effect of different odors on human emotions. We decided to choose lemon (representing "Thumbs up"), cedar (representing "Confused"), lavender (representing "Neutral"), and mint (representing "Request attention"). Lemon scents have a positive influence on the emotional states of individuals [25], mint scents can help drivers stay aware and awake [16], and lavender and cedar scents have calming and relaxing effects [38]. For relatively negative states, we did not use unpleasant scents because

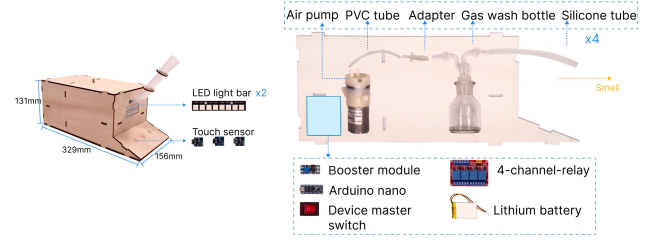


Figure 2: External (left) and internal (right) structure of an Atmospheror device.

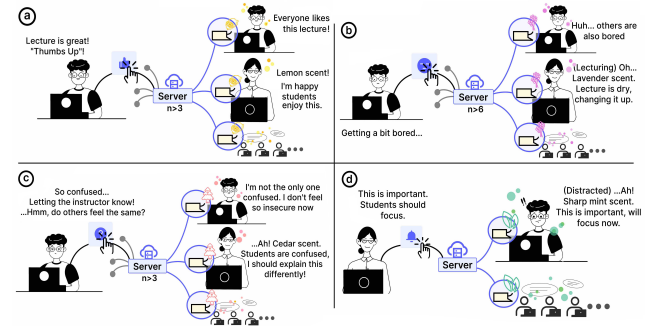


Figure 3: Atmospheror system interaction storyboards for a) Thumbs up, b) Neutral, c) Confused, and d) Request attention

they are generally used for emergency alerts and not suitable for long-term olfaction.

The Atmospheror student device has three buttons and four scent channels (lemon, lavender, cedar, and mint). The three buttons are named "Thumbs up", "Neutral", and "Confused", indicating either that students are very interested in the current content, that they feel fine/neutral regarding the current content, or that they are confused about some aspect of the current content. In contrast, the Atmospheror instructor device has two buttons and three scent channels (lemon, lavender, and cedar). The buttons are named "Next Chapter" and "Request attention". The "Next Chapter" button is a reset function, which can reset the status of student devices to the "Neutral" status. The "Request Attention" button is a reminder function that grabs student attention.

There are three key factors for scent information delivery. The first is the need to minimize latency between triggering the scent delivery and user perception of the scent. Secondly, the intensity of the scent should be suitable, and should be sufficient for user perception while not being overwhelming. Thirdly, the scent should dissipate quickly to avoid interfering with the next scent delivery [28].

3.2 Buttons, Scents, and Client Software Logic

When using the student devices, students can indicate one of three states: "Thumbs up", "Neutral", or "Confused" during class. When the number of students who are in the "Thumbs up", "Neutral", or "Confused" states goes from below a predefined threshold to above (using a rising edge trigger), all connected Atmospheror devices

(both student and instructor devices) will release the scent of lemon, lavender, or cedar respectively.

On the instructor's device, when the "Request Attention" button is pressed, all student Atmospheror devices will release the scent of mint. When the "Next Chapter" button is pressed, the status of student devices is set to the "reset" state with no scent release. In this experiment, each scent is released for 5 seconds, and the shortest interval between twice releases is 60 seconds. Students can submit the status via the button only once within 30 seconds.

4 PILOT STUDY

We conducted a pilot study using both qualitative and quantitative approaches to evaluate the effectiveness of Atmospheror (Olfactory Display Effectiveness Evaluation) and its design and implementation (UX Evaluation) from the perspectives of instructors and students. We attempt to verify the extent to which Atmospheror can help enhance social presence and interaction in synchronous online classes. Furthermore, we anticipate that user feedback will help us enhance future interaction design and experiment design.

4.1 Participants

The study involved 2 instructors (1 male and 1 female, 34 - 38 years old, $M = 36$, $SD = 2.83$) and 13 students (6 male and 7 female, 19 - 25 years old, $M = 22.8$, $SD = 2.83$). Recruited students were randomly divided into 2 equal groups of 7 and 6 students, with an instructor assigned to each. In each group, the instructor would deliver a 45-minute online lecture while using Atmospheror, during which time all participants would interact. The lecture content and participant background information is listed in Table 1.

We ensure the presentation's material would be moderately challenging to comprehend. In addition, the students were from various academic disciplines, and hence varying in-class performance and feedback were expected. All participants reported having healthy olfactory functions and no history of odor allergy and signed an informed consent form before the study.

4.2 Design and Procedure

4.2.1 Training and Memory Test. We first trained each participant for about 30 minutes and provided them with an explanation of the experiment's procedure, instructions on how to operate Atmospheror, and what each odor/button represents. Since most participants were unfamiliar with the odor notification experience, we prepared an odor memory test. First, the participants were introduced to the various odors and their corresponding meanings. Then, we would randomly play different odors and ask them to answer what the odors represented. When they got ten correct answers in a row, they were considered to have passed the test and could proceed to the next experiment stage.

4.2.2 Preparation. We provided each participant with a separate Atmospheror device. Each participant was placed in a separate and odorless space (all within the same building) and wore headphones for visual and acoustic isolation in order to simulate a realistic online classroom scenario. Tencent Meeting (VooV Meeting) was used as the video conference platform for the experiment. The hardware and software for testing were set up in advance by the researchers.

The odor delivery trigger thresholds were set by the instructor. For the threshold values of "Thumbs up", "Neutral", and "Confused", the values set by the two instructors were respectively: T1 (7 students) - 3, 6, 2; T2 (6 students) - 3, 5, 2. After the setting of thresholds was complete, the instructor would deliver a lecture that lasted about 45 minutes.

4.2.3 Data Collection. For ease of analysis, we saved screen recordings of the entire class from the instructor's perspective as MP4 files. At the end of each class, we exported device interaction logs to a database in a "timestamp: user-action" format.

To evaluate the effectiveness of Atmospheror (Olfactory Display Effectiveness Evaluation), researchers designed and drew up OCES-I (Online Class Effect Scale - Instructor version) containing 5 11-point (0-10) scale questions (e.g. concentration, student feedback, student state perception, etc.), and OCES-S (Online Class Effect Scale - Student version) questionnaire containing 5 11-point (0-10) scale questions (e.g. ability to concentrate, willingness to communicate with the instructor, etc.). The questions were explained clearly by the experiment facilitator to ensure comprehension. All subjects were assessed twice: before and after using Atmospheror. In addition to the questionnaires, we conducted semi-structured interviews with 14 participants by video conference or in-person to evaluate was of Atmospheror's design and implementation (UX Evaluation). S13 did not accept the interview for personal reasons. The interviews lasted about 20 minutes, and were audio recorded and transcribed for analysis. We first reviewed the lecture sessions through recorded videos and photos to familiarize ourselves with the real-time classroom situation. Then, the authors open-coded the transcripts and decided on preliminary topics via inductive thematic analysis [40].

5 RESULTS AND DISCUSSION

5.1 Olfactory Display Effectiveness Evaluation

We designed two Likert scales with 5 questions for each, the OCES-I and OCES-S to evaluate the effectiveness of Atmospheror. All subjects filled in a corresponding questionnaire before and after the test to compare traditional online classes to those using the Atmospheror. Due to the restricted number of instructors, the OCES-I was only used for interview induction. Table 2 displays the OCES-S items and results from students. The Shapiro-Wilk test indicated that the questionnaire data adhered to the Gaussian distribution ($p > 0.05$), thus we conducted Paired Samples T-test using R language. We calculated Mean, Standard Deviation, and Median values and compared two online class situations. Figure 4 demonstrates clearly the findings of the descriptive analysis.

In order to understand more specific impacts and supplement any aspects neglected, we conducted about 20 minutes of semi-structured interviews with 14 of the subjects (except S13). The following sections outline the quantitative and qualitative analytical results.

Both qualitative and quantitative analyses of student feedback indicated an increase in classroom engagement and concentration. Statistics from the questionnaires indicate that after using Atmospheror, "the level of student concentration" ($M = 7.92$, $p < 0.05$) and "the length of concentration" ($M = 7.69$, p

Table 1: Lecture content and participant demographics.

Lecture1				Lecture2			
Visual-audio Languages in Movies				Character Design in Animation and Digital Games			
Participant	Gender	Age	Field of Study	Participant	Gender	Age	Field of Study
T1	M	34	Animation and Film	T2	F	38	Digital Media Art
S1	F	22	Linguistics	S8	F	22	Environmental Engineering
S2	F	24	Physics	S9	F	23	Biology
S3	F	21	Industrial Design	S9	F	23	Biology
S4	M	25	Biology	S11	F	23	Digital Media Art
S5	M	25	Biomechanics	S12	M	21	Photography
S6	M	19	Automotive Engineering	S13	M	30	Art of Painting
S7	M	20	Biochemistry				

T-Instructor; S-Student

Table 2: Questions from OCES-S and results Mean, SD and Median values before and after using Atmospheror.

Online Class Effect Scale - Student version							
Question	Scale	Before			After		
		Mean	SD	Median	Mean	SD	Median
The level of your concentration	0-10	7	1.66	7	7.92	1.14	8
The length of your concentration	0-10	6.46	2.06	7	7.69	1.94	8
The workload to stay focused in class	0-10	6.46	1.95	7	6.77	1.42	7
Your fatigue in the class	0-10	5.08	2.16	6	4.38	2.27	4
The willingness to communicate with the instructor	0-10	6	1.52	6	6.69	2.05	7
0 - Little or no		10 - Extremely high					

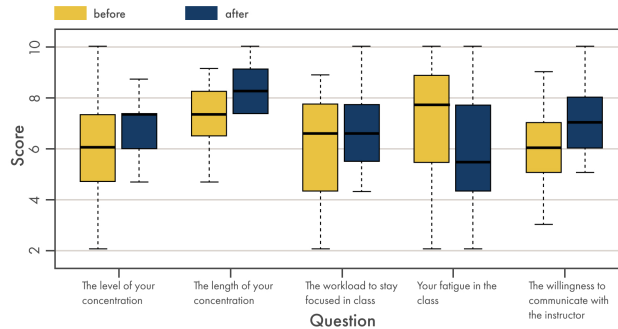


Figure 4: The results of the OCES-S questionnaire.

< 0.05) significantly increased. This optimistic conclusion reveals that Atmospheror is anticipated to have a good effect on boosting students' concentration in online classes. Even though the T-test results show that there has been no substantial change in "the willingness to communicate with instructors" ($p > 0.05$), according to the descriptive analyses, we believe there is an unavoidable tendency toward its increase, and additional iterations and tests are required. In addition, the statistical results suggest that the data validity for "the workload to stay focused in class" ($P > 0.05$) and "your fatigue in the class" ($P > 0.05$) is low and there may be no significant distinction.

The students felt that the anonymous button reduced the psychological barrier to providing feedback. Two students felt that the

status of others had little impact on their listening, but that the act of pressing the button for feedback themselves would promote greater focus and enhance their engagement (S2, S6). For example, S2 mentioned "I don't care about other people's status, it doesn't matter to me, but the stimulation I get with the equipment helps me focus." However, two students said that knowing the statuses of others would enhance their sense of connection with their classmates online (S3, S10), and S18 mentioned in the interview that "Atmospheror has increased my sense of presence and connection." In addition, students mentioned that scent can affect their moods. Since olfactory senses are rarely called upon in the classroom, many students found it novel (S5, S9, S10, S11, S12). S11 said, "The classroom has a relatively new smell, which will mobilize my attention. Even if it is used long-term, as long as you can change the smell, it will remain novel".

Instructors felt positive about Atmospheror because of the increased interactivity in the classroom. Both instructors gave Atmospheror a more positive rating (T1, T2). For example, T1 commented that "the system is a great feedback channel for students who are not very verbal" and that many students who are reluctant to communicate by text or voice are more comfortable using the anonymous feedback button (OB). Also, instructors felt that olfactory perception deepened their memory of the overall state of the classroom at each moment. This uncommon olfactory experience was more memorable to instructors than the visual and auditory feedback they used every day.

Additionally, the current study conforms to a "small class size" in teaching in terms of the number of students and teachers. Some

instructors indicated that the device might be more suitable for smaller classes because it was more helpful in locating specific problems of students. However, students opined that it was superior for larger classes since it enhanced professors' ability to observe the classroom objectively. On video conferencing services, where instructors can easily overlook text-based comments, olfactory alerts are preferable. Further, the shared scent release allows the students to be more aware of their peers. We will explore the influence of class size on the olfactory interface in the follow-up work.

5.2 UX Evaluation

As for UX evaluation, we used semi-structured interviews to investigate Atmospheror's usability, the odors chosen, the physical features of the product, and the interaction system.

5.2.1 The Recognition of Odors. We found that it is difficult for users to establish connections between certain odors and their meanings, and hence we need to make further design iterations in scent design. Both the instructors (T1, T2) and almost half of the students (S3, S4, S5, S6, S9) said it was challenging to some degree to identify the odor. Most students and instructors who had trouble identifying scents said they could not tell the difference between lavender and cedar specifically, and some of these subjects speculated that it might be because the odor was not strong enough for them. In contrast, of the four scents we used, most instructors and students felt that lemon would bring a pleasant emotional experience and could bring a sense of relaxation to the classroom (S1, S3, S4, S6, S9, S13, T2). We hypothesize that some odors that easier to distinguish represent specific emotional meanings in daily routines and common sense, whereas others do not. Therefore, when designing an olfactory interface, we should select more familiar and recognizable scents. In addition, because users have limited experience with odor as an interactive element, they must learn and practice to understand these references. The participants' adaptability must be increased by extending the duration of odor learning and long-term experiments may yield more objective and realistic outcomes.

5.2.2 Hardware Implementation. The system uses four air pumps and four gas wash bottles to deliver odorous air in real-time. However, these components are large and lead to the device taking up much space overall. Some subjects also indicated that they had to keep their faces close to the device to ensure that they received the odor information promptly. This drawback brought about some feedback during the user interviews regarding how there were some aspects of feeling "unnatural". Another hardware limitation is the noise generated by the air pump, which is difficult to ignore completely. Although we had used acoustic foam to fill the device's interior and required all subjects to wear headphones, a few subjects reported that they could still hear the air pump noise. We plan to use a variety of methods to minimize the device size and to control noise in future iterations, including but not limited to 1) using a smaller customized odor source storage container, 2) using a smaller, quieter air pump, 3) integrating the control circuit, 4) redesigning the internal device layout and structure, and

5) choosing better sound insulation materials. For subsequent experimental improvements, we would 1) equip participants with noise-canceling headphones and 2) extend the output pipe to hide the odor-generation device further.

5.2.3 Interaction System. This on-going project investigates the effectiveness of an olfactory interface in synchronous online classes and provides stimulus and inspiration for future development. Consequently, we endeavor to reduce the user system's complexity. In order to maintain the logical rationality of the students' responses to odor triggers, a time limit has been imposed on this experiment. Each scent is released for a duration of 5 seconds, with a minimum interval of 60 seconds between releases. Students can only submit their status using the button once every 30 seconds in order to prevent them from intentionally disrupting the order. In addition, we must determine the odor threshold, which requires a voting procedure. Nonetheless, we believe this should be reconsidered. In addition, the design of the student voting cooling period merits further refinement. A superior interaction system would unquestionably require more attention to detail and deliberation.

6 CONCLUSION AND FUTURE WORK

In this project, we aim to explore the usage of the olfactory modality in synchronous online learning. We designed and implemented Atmospheror, a system that uses an olfactory display to give instructors greater awareness of student status, creates a shared "ambient" atmosphere in the virtual classroom, and encourages students to engage actively in communication. We then conducted a pilot study that verified the usability of Atmospheror and explored the impact it could have on synchronous online classes.

Student responses suggests that there was an enhancement in classroom engagement and concentration. The anonymous button and olfactory interface decreased the psychological barrier for students to submit feedback. Moreover, odor could impact moods and provide a novel sensory experience. Instructors viewed Atmospheror favorably due to the enhanced interactivity within the classroom. Additionally, olfactory sensitivity strengthened their recollection of the classroom's status at any given time.

The preliminary results of this pilot study cannot be considered precise. We understand some limitations, such as a flawed control group setting and insufficient device size. However, we still received some insightful findings and positive participant feedback (both from the instructor and the student perspectives), which inspired the direction of subsequent iterations and studies. We will increase efforts in the following three aspects: 1) We expect to select more identifiable odors and design a pre-training process for users to form odor-semantic memories in a long-term experiment. 2) At the same time, we will develop the hardware to be more user-friendly and add customized settings in the client software. 3) Moreover, we would further investigate the interaction mechanisms in order to support a better multi-person real-time system.

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